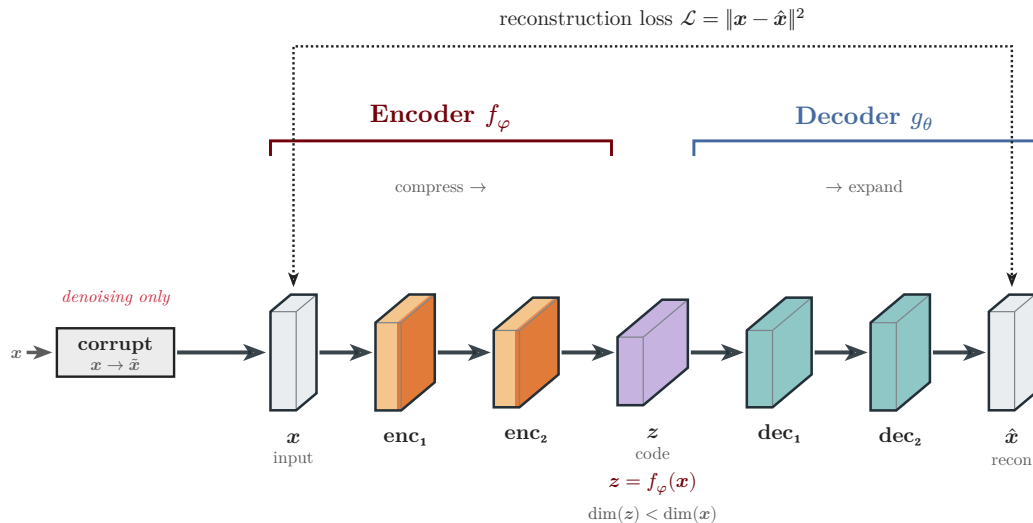




The four family members share one scaffold — encoder  $\rightarrow$  code  $\rightarrow$  decoder — and differ in the colored add-on

### Undercomplete

bottleneck dim  $<$  input dim;  
compression is the only regularizer

objective:  
 $\|\mathbf{x} - \hat{\mathbf{x}}\|^2$

### Denoising

corrupt input  $\mathbf{x} \rightarrow \tilde{\mathbf{x}}$ ,  
reconstruct the clean  $\mathbf{x}$

objective:  
 $\|\mathbf{x} - g_\theta(f_\varphi(\tilde{\mathbf{x}}))\|^2$

### Sparse

penalize code activations  $\Omega(z)$   
so few units fire per input

objective:  
 $\|\mathbf{x} - \hat{\mathbf{x}}\|^2 + \lambda\Omega(z)$

### Masked

hide a subset of input  
tokens / patches, predict them

objective:  
 $\sum_{i \in \mathcal{M}} \|\mathbf{x}_i - \hat{\mathbf{x}}_i\|^2$